Unit 9: Worked Examples

1. Identify if the following questions are statistical or non-statistical.

   a. How old is 'The Mona Lisa'? Non - statistical

   b. What was the score of students on the Florida State Assessment? Statistical

   c. Is this a statistical question? No - non statistical

   d. How many siblings do your classmates have? Statistical

2. A marine biologist is studying turtles and wants to calculate the average length of time a Red-Eared Slider, a popular breed of turtle, can remain underwater. The time, in minutes, that ten Red-Eared Sliders remain underwater are,

   65, 310, 215, 400, 250, 195, 330, 45, 165, 430.

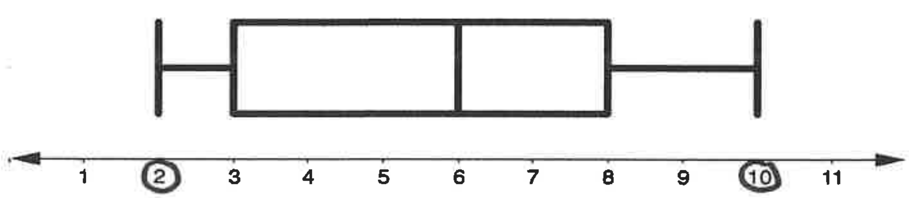
   Calculate the mean, medium, mode and range of the date set.

   45, 65, 165, 195, 215, 250, 310, 330, 400, 430

   sum = 2405

   mean = $\frac{2405}{10}$ = 240.5 median = 232.5 mode = no mode range = $\begin{array}{r} 430 \\ -\ 45 \\ \hline 385 \end{array}$

3. A soccer coach recorded the number of freekicks his team earned per game over the course of a year. The data set below represents her findings.



   a. Identify the least and most amount of freekicks awarded each game.

   least = 2 most = 10

   b. What was the range of freekicks awarded each game?

   10 - 2 = 8

   c. What is the median number of freekicks awarded each game?

   6

   d. What is the interquartile range?

   8 - 3 = 5